



Nutri-Score Project

Make healthy choices the easy choices

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Task 4.1:

Data Collection & Validation



24 food categories

Cookies, bread, beverages, dairy, etc...

product_name	brands	categories	energy_100g	saturated_fat_100g	sugars_100g	salt_100g	proteins_100g	fiber_100g	fvl_percent	nutri_score_value	nutri_score_label	green_score_value	green_score_label	url
14 Maxi Tranches complet sans sucres ajoutés	Jacquet	pain	1026	0.3	3.8	1.2	8.7	6.6	0	2	B	83	A	https://world.openfoodfacts.org/product/3029330022428

+1,000 products target

Next step: Find the possible error labeling products

Task 4.2: Nutri-score

Score range	Class	Color
[-17;0]	A	Dark green
[1;2]	B	Light green
[3,10]	C	Yellow
[11;18]	D	Light orange
[19;55]	E	Red or Dark orange

Table 3: 5-color nutritional classification



Function	Description
Negative points (N)	Σ energy, saturated fat, sugars, salt
Positive points (P)	Σ proteins, fiber, fruit/vegetables $N \geq 11$ AND fruit/veg $\leq 80\%$ $\rightarrow$ proteins NOT counted
Score (computed_score)	$N - P$

product_name	brands	categories	energy_100g	saturated_fat_100g	sugars_100g	nutri_score_value	nutri_score_label	computed_score	computed_class	N	P
14 Maxi Tranches complet sans sucres ajoutés	Jacquet	pain	1026	0.3	3.8	2	B	2	B	9	7

Task 4.2: Nutri-score - Accessible Live on [Hugging Face](https://huggingface.co/spaces/NimaKL/Nutri-Score)

About Nutri-Score

The Nutri-Score is a nutrition label that converts the nutritional value of products into a simple code consisting of 5 letters (A to E), each with its own color.

How it works:

- A (Dark Green): Best nutritional quality
- B (Light Green): Good quality
- C (Yellow): Average quality
- D (Orange): Poor quality
- E (Red): Worst quality

Calculation:

- Negative Points (N): Energy, sugars, saturated fat, salt
- Positive Points (P): Proteins, fiber, fruits/vegetables
- Score = N - P

Lower scores = Better nutrition

Example Products

1. Low-Fat Plain Yogurt – Nutri-Score A

Nutrients per 100g

- Energy: 250 kJ
- Saturated Fat: 0.2 g
- Sugars: 3.5 g
- Salt: 0.01 g
- Protein: 5.5 g
- Fiber: 0 g

Nutri-Score Calculator

Calculate the nutritional quality of food products

Input Nutritional Values (per 100g/100ml)

Nutrients to Limit

Energy (kJ) *

1500.00

Saturated Fat (g) *

5.00

Sugar (g) *

10.00

Salt (g) *

1.00

Nutrients to Favor

Protein (g) *

8.00

Fiber (g) *

3.00

Fruit/Vegetables/Nuts (%)

20.00

Calculate Nutri-Score

Results

Nutri-Score: D

Nutri-Score Scale

A

B

C

D

E

Score Breakdown

Negative Points (N)

14

Positive Points (P)

0

Final Score

14

Interpretation

Poor nutritional quality. Limit consumption of this product.

View Calculation Details

Task 4.3: ELECTRE TRI Results & Comparison

Method: Pessimistic and Optimistic Majority Sorting

Classes Defined (A' to E'):

- A': Excellent nutritional & environmental profile
- B': Good profile
- C': Average profile
- D': Poor profile
- E': Very poor profile

Green-Score (30%): Environmental impact increasingly important for food decisions; reflects EU policy priorities

Balanced approach: Negative criteria (35% total) vs Positive criteria (35% total) ensures neither dominates

Salt (5%): Lower than other negative criteria because health impact requires long-term cumulative exposure

Proteins/Fiber (12% each): Essential nutrients often deficient in modern diets

Criteria Weights ([electre_tri_weights.csv](#)) - JUSTIFIED:

Green-Score:	30%	← Highest weight (environmental priority)
Proteins:	12%	← Key nutrient to favor
Fiber:	12%	← Key nutrient to favor
Fruits/Veg:	11%	← Positive component
Energy:	10%	← Major negative component
Saturated Fat:	10%	← Major negative component
Sugars:	10%	← Major negative component
Salt:	5%	← Lower health impact

TOTAL:	100%
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Chosen quantiles

$\pi_1=20\%$, $\pi_2=35\%$, $\pi_3=50\%$,
 $\pi_4=65\%$, $\pi_5=80\%$, $\pi_6=90\%$.

Task 4.3: ELECTRE TRI Results & Comparison

Limiting Profiles ([electre_tri_limiting_profiles.csv](#)):

- **Method:** Statistical quantile approach based on database distribution
- **6 Profiles:** π_1 (lower bound) $\rightarrow \pi_6$ (upper bound)
- Profiles define boundaries between classes A'/B'/C'/D'/E'

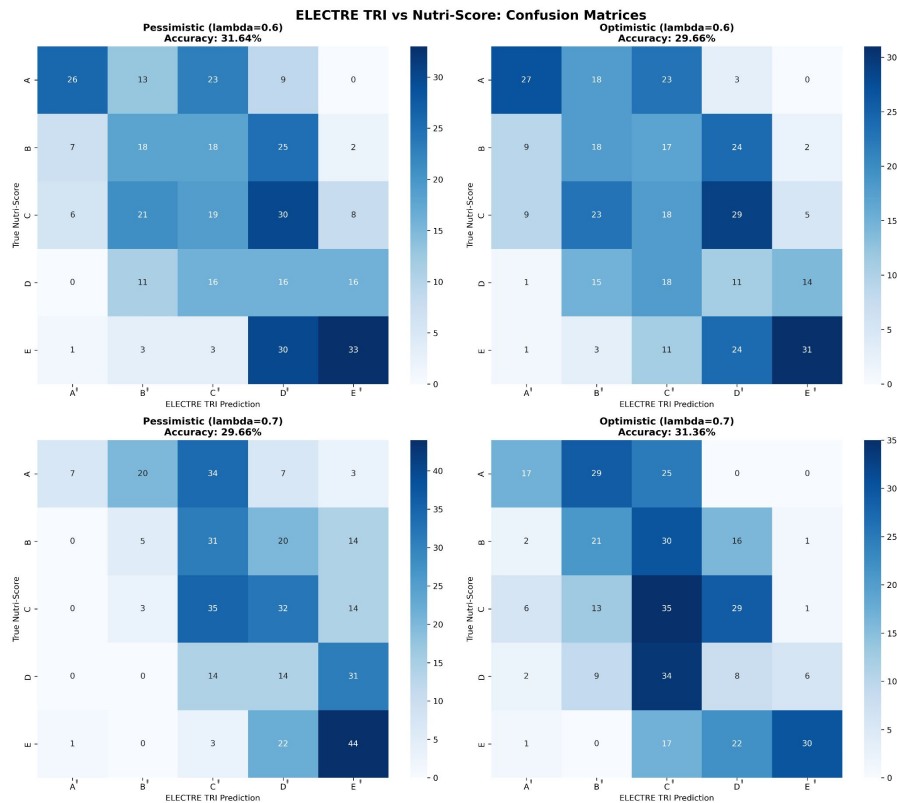
Implementation:

- Tested with $\lambda = 0.6$ and $\lambda = 0.7$ majority thresholds
- Results exported:
[electre_tri_results_lambda_0_6.csv](#)
[electre_tri_results_lambda_0_7.csv](#)

green_score	nutri_score	ELECTRE_ Pessimistic_06	ELECTRE_ Optimistic_06
A	A	C	B
A	B	B	B
B	B	B	B
C	C	D	D

green_score	nutri_score	ELECTRE_ Pessimistic_07	ELECTRE_ Optimistic_07
A	A	C	B
A	B	C	B
B	B	C	C
C	C	D	D

Task 4.3: ELECTRE TRI Results & Comparison



Accuracy ~30%:

- Only 30% of products get the same class in both methods
- This shows the two methods are **fundamentally different**

Systematic Downward Shift:

- **Nutri-A** products → mostly ELECTRE **B, C** (1-2 classes lower)
- **Nutri-B** products → mostly ELECTRE **C, D** (1-2 classes lower)
- **Nutri-E** products → ELECTRE **D, E** (agreement on bad products!)

$\lambda=0.7$ is MUCH harsher:

- Pessimistic $\lambda=0.7$: Almost NO products reach A' or B'
- Most products pile into C
- Optimistic $\lambda=0.7$: Slightly better but still conservative

Task 4.3: ELECTRE TRI Results & Comparison

Nutri-Score distribution (blue):

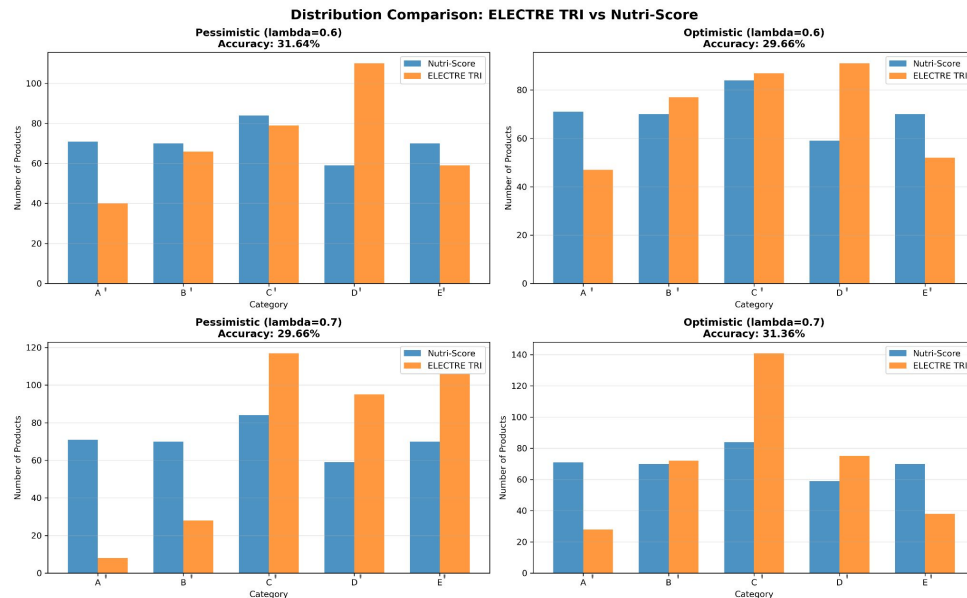
- **Balanced** across all classes
 - Many A's (~71), B's (~70), C's (~84)

ELECTRE TRI distribution (orange):

- **Pessimistic $\lambda=0.6$:**
 - Few A's (~40)
 - many D's (~108)
- **Pessimistic $\lambda=0.7$:**
 - Even fewer A's (~8),
 - MASSIVE C spike (~117)
- **Optimistic $\lambda=0.6/0.7$:** Better but still C-D heavy

Interpretation:

- ELECTRE TRI creates a **"middle-heavy"** distribution (lots of C's and D's)
- Very few products reach the top classes (A', B')
- This is the **non-compensatory effect + profile calibration**



Task 4.4: Weighted Sum Model

Model: Combined score integrating all 8 criteria

Approach:

- Normalize all criteria to [0, 1] scale
- Apply weights to compute aggregate score
- Define threshold ranges for classes A' to E'

Justification:

- Using identical weights allows direct comparison with ELECTRE TRI
- Tests whether simpler aggregation method yields similar classifications
- Weighted sum assumes compensatory tradeoffs (poor nutrition can be offset by good environmental score)

Key Difference from ELECTRE TRI:

- Weighted sum: Full compensation allowed between criteria
- ELECTRE TRI: Non-compensatory (concordance/discordance logic prevents full compensation)

Weights Used (from `electre_tri_weights.csv`):

Same as ELECTRE TRI for comparability:

- Green-Score: 30%
- Proteins: 12%, Fiber: 12%, Fruits/Veg: 11%
- Energy: 10%, Sat Fat: 10%, Sugars: 10%, Salt: 5%

Task 4.4: Weighted Sum Model

Nearly **perfectly balanced** across A–E (≈ 70 each).

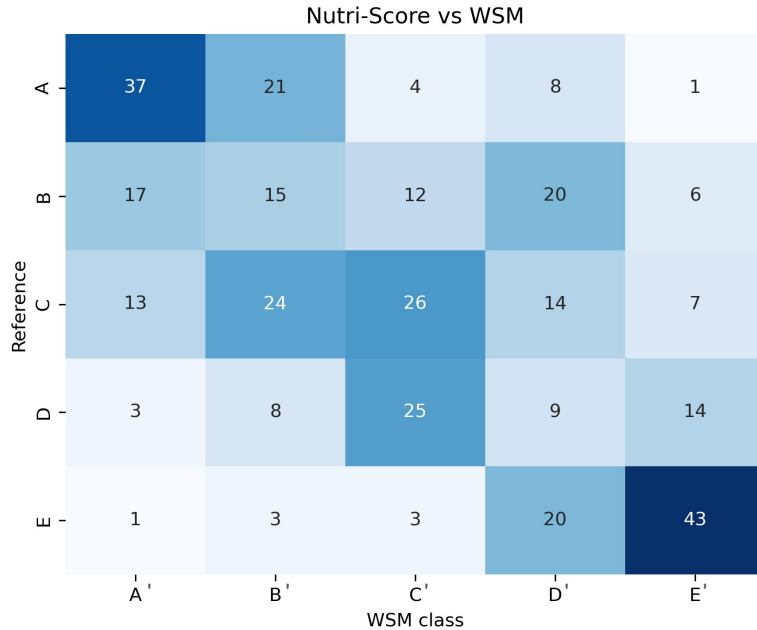
Result of **uniform score thresholds** in the linear weighted-sum model.

Contrasts with ELECTRE TRI's **C–D heavy distribution**.

Shows how **compensatory scoring smooths differences** between products.

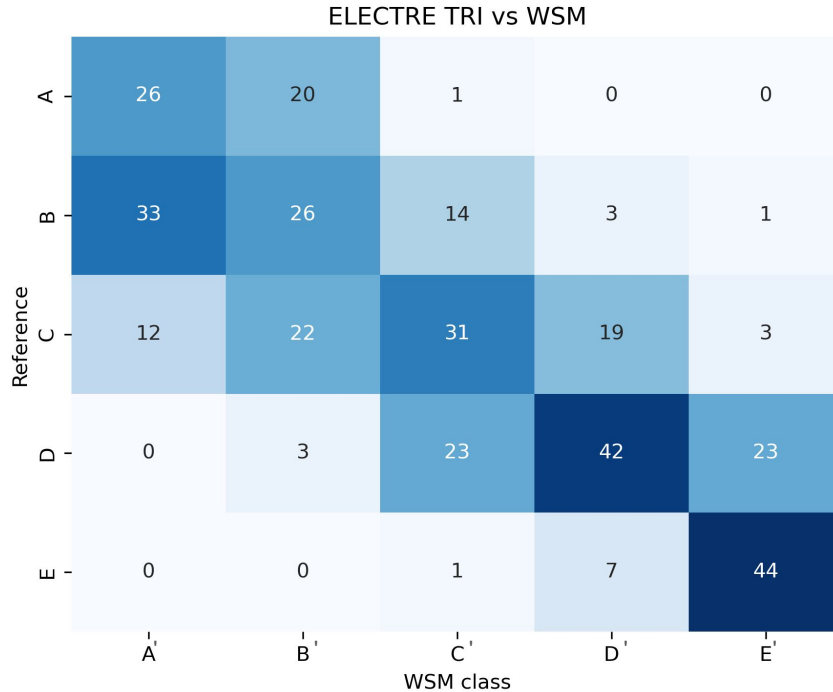


Task 4.4: Weighted Sum Model



- Strong agreement only for **A** and **E**.
- Middle classes (B–D) show **large dispersion**, highlighting methodological differences.
- Nutri-A values often downgraded to **WSM-B/C** → WSM penalizes certain criteria differently.
- Confirms that **WSM and Nutri-Score reflect distinct scoring philosophies**.

Task 4.4: Weighted Sum Model



- Clear **diagonal dominance** → good agreement for C, D, E classes.
- **A and B classes differ strongly:**
 - Many ELECTRE-A/B products become **WSM-B/C**.
 - Shows that **WSM is more compensatory**, allowing products with strong environmental scores to move upward.
- **High agreement for low-performing products:**
 - ELECTRE-D/E → mostly WSM-D/E.
- **Overall pattern:**
 - ELECTRE TRI classifies products more conservatively (middle-heavy).
 - WSM produces smoother, more continuous differentiation.

Key takeaway:

Methods diverge mainly on healthier products (A/B), but align on poor ones (D/E).

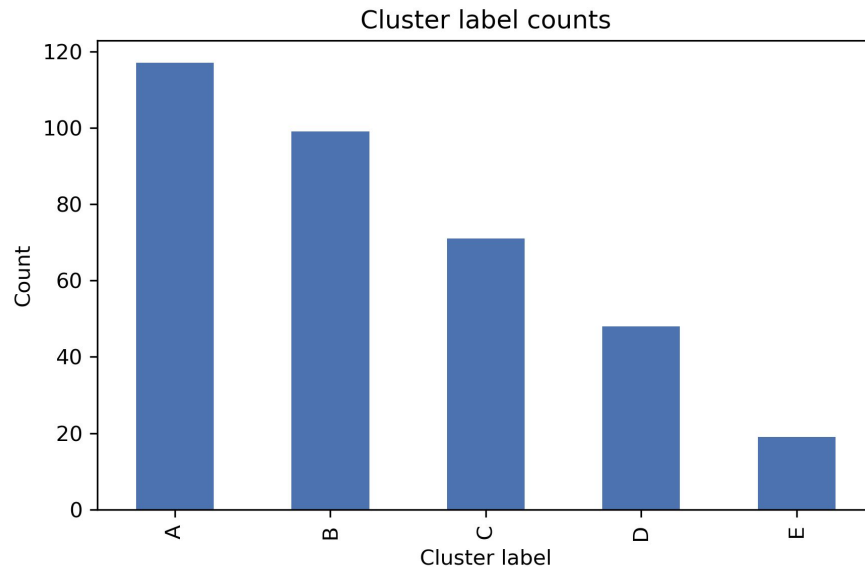
Task 4.5: K-Means

Cluster Distribution

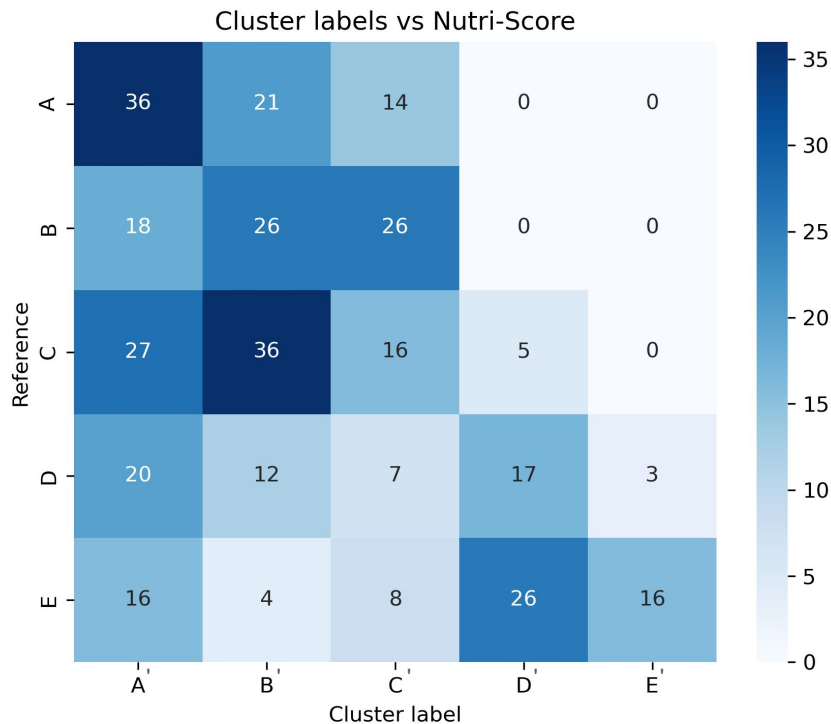
- Strongly unbalanced clusters:
 - **Cluster A/B (best groups)** very large (≈ 115 and ≈ 98 products).
 - **Cluster C/D** moderate.
 - **Cluster E** small (≈ 19 products).
- Reflects real market composition: many products are “acceptable,” few are extremely poor.

Key takeaway:

K-means naturally produces a pyramidal structure: many average products, few extreme outliers (good or bad).



Task 4.5: K-Means



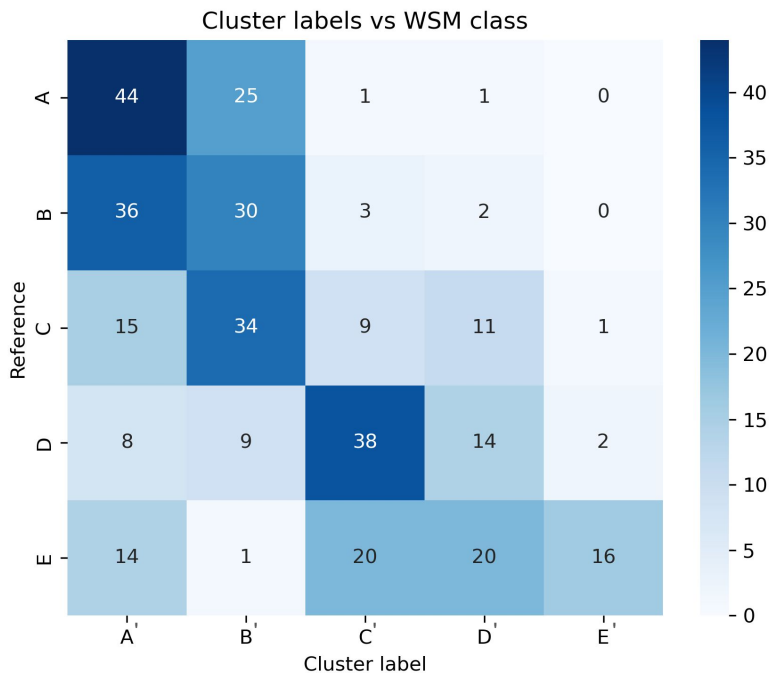
Clustering vs Nutri-Score

- Much more mixing than with WSM → clusters integrate additional dimensions (including Green-Score) that Nutri-Score ignores.
- Nutri-A/B/C spread across clusters A/B/C → clusters reflect **multi-criteria similarity**, not nutritional strictness.
- Cluster D/E gather many Nutri-Score D/E products → agreement strongest for low-quality items.

Key takeaway:

Clusters diverge from Nutri-Score, confirming that adding Green-Score + other criteria reshapes the product landscape.

Task 4.5: K-Means



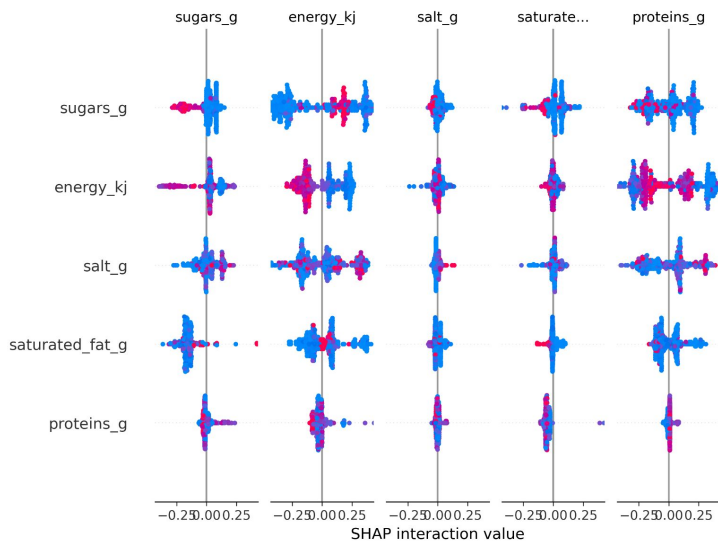
Clustering vs WSM

- Reasonable alignment with WSM for **A, B, and D**, but clusters show **stronger separation** than WSM.
- **Cluster A/B** capture most WSM A/B products → clusters identify a *healthy–moderate* axis similar to WSM.
- **Cluster C/D** capture mid-range products with mixed criteria.
- Cluster E isolates a small group of **worst-performing products**.

Key takeaway:

K-means recreates a structure close to WSM but with clearer group boundaries and a more distinct “bad products” cluster.

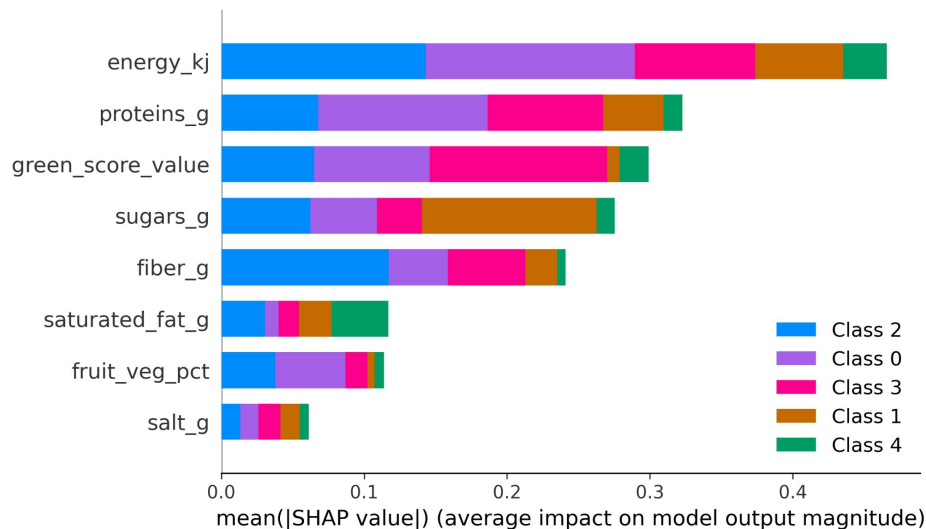
Task 4.5: K-Means



Pairwise SHAP effects show that cluster assignment is mainly driven by simple nutritional patterns.

- Strongest interactions: **energy–sugars, sugars–sat fat, energy–proteins.**
- High sugars/energy → push products to *worse* clusters.
- Higher proteins → push to *better* clusters.

Takeaway: Clusters reflect intuitive nutrition relationships; no complex nonlinear behavior.

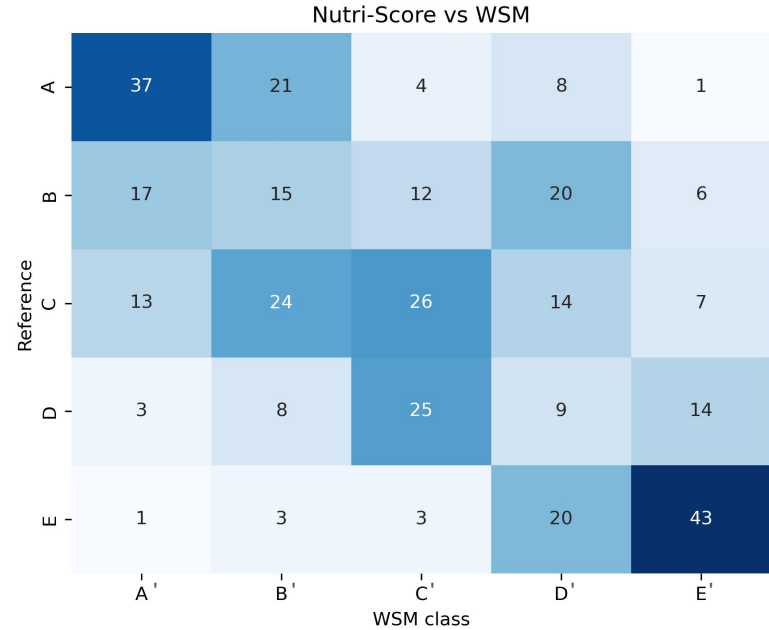
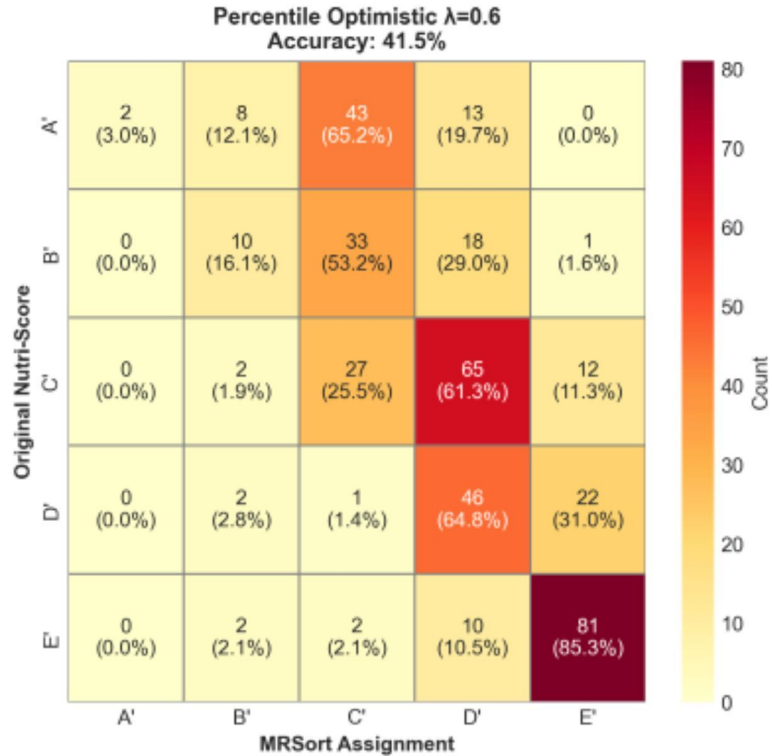


Top drivers of the new scoring system:

1. **Energy**
2. **Proteins**
3. **Green-Score**
4. **Sugars, Fiber**

Takeaway: The model balances **nutrition + environmental impact**, with energy/sugars lowering scores and proteins/Green-Score improving them.

Task 4.6: Compare with another Group





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